



华南理工大学
South China University of Technology

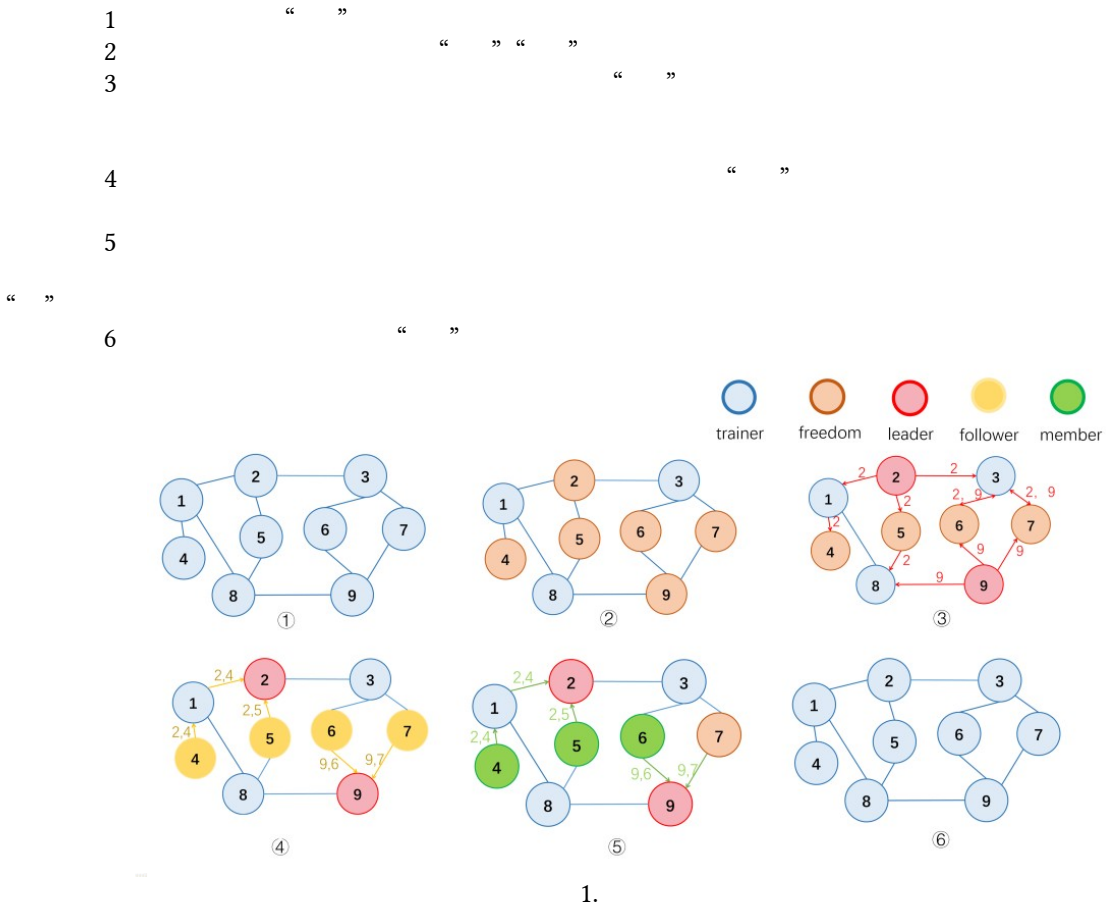
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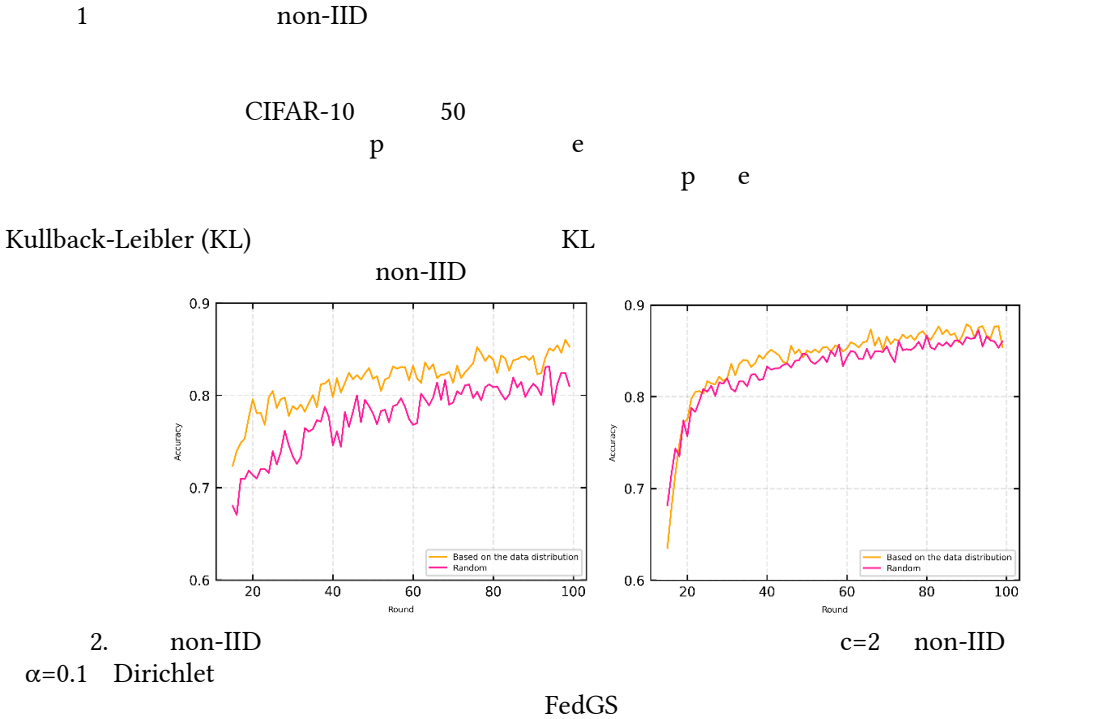
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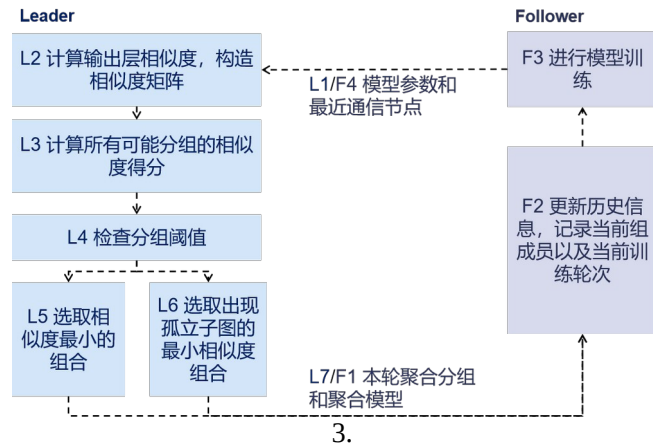
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1 IoT E-tree FedCH CASP GossipFL non-IID (AMAS) AMAS 2 ① ②	
1 $D \triangleq \sum_{i=1}^N D_i$$f(x_j, y_j, \omega_i)$$hr(i) \subset \{1, 2, \dots, N\} \setminus \{i\}$ $\min_{\omega, \{\omega_i\}_{i=1}^N} \sum_{i=1}^N F_i(\omega_i)$$s.t. \quad \omega_i = \omega, \forall i.$ 2 2.1	V_i D_i ω_i $F_i(\omega_i) \triangleq \frac{1}{ D_i } \sum_{j \in D_i} f_j(\omega_i)$ $f_j(\omega_i)$ j x_j y_j D_i i $i \ j$ B_{ij} $i \in N$ n



3.2





3

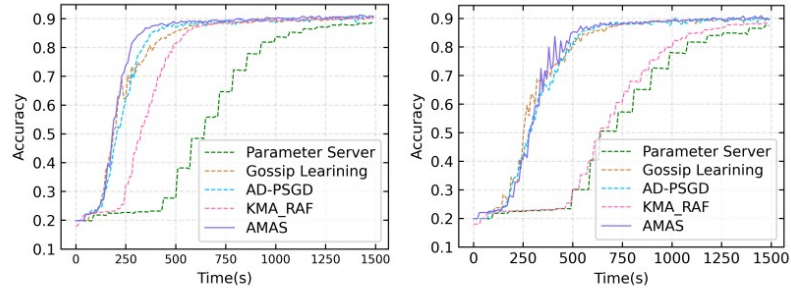
3.1

1	(CNN)	Fashion-MNIST	CIFAR-10	SVHN	CNN	3
2	SGD	32	0.01			
2	non-IID		C=K		k	
3	$pk \sim \text{Dir}(\alpha)$	Dirichlet	CPU	α	2-6	CPU
	5	200				

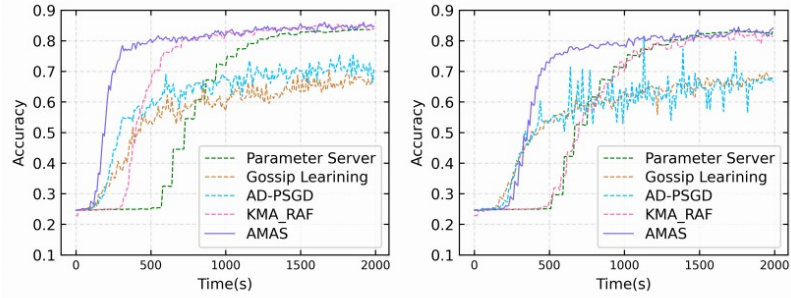
3.2

AMAS	AMAS	IID	non-IID	AMAS	AD-PSGD	Gossip Learning
				non-IID		
AMAS						

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(a) IID-data without dynamic heterogeneity. (b) IID-data with dynamic heterogeneity.



(c) non-IID data without dynamic heterogeneity. (d) non-IID data with dynamic heterogeneity.

4.

IID

IID

			Parameter Server	Gossip Learning	AD-PSGD	KMA.RAF	AMAS
IID	deterministic	Accuracy(%)	88.61	90.99	90.87	90.22	91.27
		Speed(s)	1330	620	490	670	420
	non-deterministic	Accuracy(%)	87.53	90.11	90.38	88.49	90.93
		Speed(s)	1470	690	620	1270	540
non-IID	deterministic	Accuracy(%)	83.88	70.26	75.68	85.49	86.05
		Speed(s)	940	1690	1150	530	250
	non-deterministic	Accuracy(%)	83.56	69.68	80.86	82.87	84.48
		Speed(s)	930	1940	640	970	450

1. AMAS

IID

1500

non-IID

2000

70% 88% 69% 87%

Fashion-MNIST CIFAR-10 SVHN

pk~Dir(0.1) C=2 C=3

KL

KL

2

Fashion-MNIST CIFAR-10 SVHN

1500

50% 80% 70%

	Fashion-MNIST		CIFAR-10		SVHN	
	selected	unselected	selected	unselected	selected	unselected
KL divergence	0.4438	0.4676	0.2531	0.3288	0.1998	0.2341
Cosine Similarity	0.7259	0.7167	0.8493	0.8168	0.8533	0.8396
Accuracy(%)	55.73	53.39	88.28	88.36	78.90	76.28
Speed(s)	770	1090	330	380	560	710

2.

(2-3)

(4-5)

:1500s

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50% 80% 70%

4

4.1

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4.2

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②

non-IID

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